

Factorial Moments in Two Variables: Exact Leaf Transport and Trace Orthogonality

Christopher D. Long
galizur@gmail.com

7 October 2026

Abstract

Let $\mathcal{L}(x^a y^b) = a!b!$. We give a proof of the two-variable Factorial Conjecture using the explicitly stated single-integral nonvanishing theorem of EIT I for the constant projective leading case. For a nonconstant projective leading polynomial, an exact radial Mittag–Leffler transform produces vanishing leaf germs containing both a pole density and a Volterra correction. The corrections admit continuation with the complete integration histories retained; at a nonzero terminal critical value they have no negative Puiseux coefficients. Polynomial path-module theory therefore forces the weighted leading coefficient to annihilate the top rational monodromy component. The unweighted inverse-derivative vector belongs to that component, by a zero-trace identity on every proper composition block. Conjugate pairing and Lindemann–Weierstrass exclude all nonzero critical values. Endpoint logarithms dispose of the remaining pure powers. All factorial-specific constructions are proved here. Appendices give exact continuation examples and effective differential-system certificates not used in the main proof.

1 The statement and the inputs

Define the complex-linear functional $\mathcal{L} : \mathbb{C}[x, y] \rightarrow \mathbb{C}$ by $\mathcal{L}(x^a y^b) = a!b!$ for $a, b \geq 0$. The Factorial Conjecture was introduced by van den Essen, Wright and Zhao [2]. Liu and Sun proved the homogeneous two-variable case [3]. The present argument treats arbitrary lower homogeneous terms.

For $0 \neq f \in \mathbb{C}[x, y]$ of total degree $D \geq 1$, write

$$P(t, r) = f(rt, r(1-t)) = \sum_{j=0}^D A_j(t) r^{D-j}, \quad \deg A_j \leq D-j, \quad A = A_0. \quad (1.1)$$

Thus $A(t) = f_D(t, 1-t)$ is nonzero. Its degree d can be smaller than D . We call the cases $d > 0$ and $d = 0$ *nonflat* and *flat*, respectively.

Theorem 1.1 (Nonflat nonvanishing). *If $0 \neq f \in \mathbb{C}[x, y]$ has nonconstant A , then $\mathcal{L}(f^n) \neq 0$ for some $n \geq 1$, and in fact for infinitely many n .*

Theorem 1.2 (EIT I: the external nonvanishing input). *Let $q \in \overline{\mathbb{Q}}[t]$ be nonconstant, let $a, b \in \overline{\mathbb{Q}}$ be distinct, and let $\xi \in \overline{\mathbb{Q}}^\times$. Then*

$$\int_a^b e^{\xi q(t)} dt \neq 0.$$

This is the nonvanishing part of the single-integral corollary of EIT I [5, Corollary cor:single]; that result also asserts transcendence. The integral can be taken along any finite path, since its integrand is entire. No extension to arbitrary amplitudes is being used.

Corollary 1.3 (FC(2), using Theorem 1.2). *For every $f \in \mathbb{C}[x, y]$,*

$$(\mathcal{L}(f^n) = 0 \text{ for every } n \geq 1) \implies f = 0.$$

Equivalently, every nonzero f has infinitely many nonzero factorial moments.

Direct dependencies and scope. The nonflat proof uses Lindemann–Weierstrass and the top-component irreducibility theorem for rational permutation modules of groups containing a full cycle [4, Theorem 3.1]. We state precisely the part of that representation theorem needed below, and prove the endpoint averaging, planar separation, and resulting path-module containment in our notation. We do not reprove the underlying Schur-ring classification. The flat step uses only Theorem 1.2. Its source is a separate research manuscript whose proof uses E -function lifting and Fourier–Laplace theory; those are transitive, not additional direct, inputs here. Apart from standard complex analysis, algebraic inverse branches, and the Nullstellensatz and Lüroth theorems, all remaining arguments are included.

Manuscript status. This is an unrefereed research manuscript, not a Lean formalization. The EIT dependence of Corollary 1.3 is explicit; the nonflat argument is independent of EIT. The exact checks accompanying the source verify particular identities, not the universal theorem.

2 Normalization and the exact kernel

Lemma 2.1 (Algebraic specialization). *A complex counterexample with specified total degree D and specified positive degree d of A would give a counterexample over $\overline{\mathbb{Q}}$ with the same degrees. The same assertion holds with the normalized flat top $A = 1$.*

Proof. Use the finitely many coefficients of a polynomial of degree at most D as variables. All equations $\mathcal{L}(f^n) = 0$ have rational coefficients. Impose the linear conditions specifying $\deg A = d$, and adjoin the inverse of its degree- d coefficient. In the flat case impose $A = 1$, after scaling the proposed counterexample by its nonzero constant leading value. The assumed complex point makes this ideal proper. Noetherianity and the Nullstellensatz over $\overline{\mathbb{Q}}$ give an algebraic point satisfying every moment equation. The inverse variable preserves d and D : restriction of a homogeneous form to $x + y = 1$ is injective. Neither the monodromy group nor the decomposition lattice has to survive specialization, since the contradiction is uniform in both. $\square$

Set

$$M_n = \frac{\mathcal{L}(f^n)}{\Gamma(Dn + 2)}, \quad \widehat{G}(w) = \frac{1}{w} \sum_{n \geq 0} M_n w^{-n}, \quad (2.1)$$

$$E_{D,2}(z) = \sum_{n \geq 0} \frac{z^n}{\Gamma(Dn + 2)}, \quad \mathcal{C}(w, t) = \frac{1}{w} \int_0^\infty e^{-r} r E_{D,2}\left(\frac{P(t, r)}{w}\right) dr. \quad (2.2)$$

Termwise integration of a polynomial gives

$$\mathcal{L}(f^n) = \int_0^\infty \int_0^\infty e^{-x-y} f(x, y)^n dx dy = \int_0^1 \int_0^\infty e^{-r} r P(t, r)^n dr dt.$$

The last equality uses the Jacobian r of $x = rt$, $y = r(1 - t)$. For sufficiently large $|w|$ the resulting generating integral is normally convergent, and

$$\widehat{G}(w) = \int_0^1 \mathcal{C}(w, t) dt. \quad (2.3)$$

Indeed, the root filter below bounds $E_{D,2}(z)$ by a polynomial factor times $e^{|z|^{1/D}}$, and $|P(t, r)|^{1/D} \leq C(1+r)$ on a compact t -set. The same estimate justifies summation under the integral. All positive moments vanish exactly when $\widehat{G}(w) = 1/w$ near infinity.

Put $V = \{t : A(t) \neq 0\}$. We need a coefficient on a finite cover, not a single-valued logarithmic coefficient on the entire base.

Lemma 2.2 (Exact local kernel and a continued coefficient). *Near $w = A(t) \neq 0$, a continued germ of the kernel has the form*

$$\mathcal{C}(w, t) = \frac{E(t)}{w - A(t)} + B_0(w, t) \log(w - A(t)) + H(w, t), \quad E(t) = \exp\left(\frac{A_1(t)}{DA(t)}\right), \quad (2.4)$$

where B_0, H are locally jointly holomorphic. The residue is unchanged by continuation. For fixed $t \in V$, the only possible finite exceptional values for continuation in w are $0, A(t)$, locally uniformly in t .

The distinguished diagonal germ of B_0 extends to a holomorphic function $\mathcal{B}$ on

$$\mathcal{X} = \{(w, t, \rho) : w \neq 0, t \in V, \rho^D = A(t)/w\}. \quad (2.5)$$

It is given by

$$\mathcal{B}(w, t, \rho) = \frac{\rho^{D-1}}{DA(t)} \sum_{m \geq 1} \frac{(-1)^m F_m(t)}{(m-1)!} (1-\rho)^{m-1}, \quad (2.6)$$

where F_m are defined in (2.8). The series is entire in $1-\rho$, locally uniformly on V . The initial diagonal germ is the restriction with ρ near 1.

Proof. Let $\omega = e^{2\pi i/D}$. Expansion of the exponentials gives the exact root filter

$$E_{D,2}(X^D) = \frac{1}{DX} \sum_{k=0}^{D-1} \omega^{-k} e^{\omega^k X} - C_{D,1}(X), \quad C_{D,1} = \begin{cases} X^{-1}, & D = 1, \\ 0, & D \geq 2. \end{cases} \quad (2.7)$$

Apply it only after splitting the radial integral at a sufficiently large R . Leave the compact radial part in its entire Mittag-Leffler form.

On V , the exterior root with leading term r and its inverse are

$$\Phi_t(r) = (P(t, r)/A(t))^{1/D} = r + p_1(t) + O(r^{-1}), \quad \chi(t, Y) = Y - p_1(t) + O(Y^{-1}), \quad p_1 = \frac{A_1}{DA}.$$

The analytic inverse theorem at infinity gives convergent Laurent series, uniform on exterior disks for compact subsets of V . The chosen leading terms specify them uniquely; their coefficients are single-valued and holomorphic on V . Define

$$F(t, Y) = e^{Y-\chi(t, Y)} \frac{\chi(t, Y) \chi_Y(t, Y)}{Y} = \sum_{m \geq 0} F_m(t) Y^{-m}. \quad (2.8)$$

Here $F_0 = E$ and locally $|F_m(t)| \leq CR_0^m$ for fixed C, R_0 . With $\rho = (A(t)/w)^{1/D}$ near 1, the summand singular at this diagonal is

$$\frac{\rho^{D-1}}{DA(t)} \int_{\Phi_t([R, \infty))} e^{-(1-\rho)Y} F(t, Y) dY. \quad (2.9)$$

In $\Re(1-\rho) > 0$ the exterior image ray can be moved to a common positive ray. The far connector tends to zero exponentially, and the finite connector is entire in $1-\rho$, locally uniformly in t .

Choose a starting point $R_1 > R_0$ on that ray and put

$$J_m(\eta) = \int_{R_1}^{\infty} e^{-\eta Y} Y^{-m} dY.$$

Direct integration and integration by parts give

$$\begin{aligned} J_0 &= e^{-\eta R_1} / \eta, \\ J_1 &= -\log \eta - \gamma - \log R_1 - \sum_{j \geq 1} \frac{(-\eta R_1)^j}{j j!}, \\ J_{m+1} &= \frac{e^{-\eta R_1} R_1^{-m} - \eta J_m}{m} \quad (m \geq 1). \end{aligned}$$

The logarithmic coefficient of J_m is $(-1)^m \eta^{m-1} / (m-1)!$. For $m \geq 1$ its remaining part is entire; on compact η -sets the recurrence bounds it by a fixed constant times $R_1^{1-m} + L^{m-1} / (m-1)!$. Combining this with the geometric bound on F_m proves normal convergence of both continued series. Thus the tail is

$$\frac{E(t)}{\eta} + \left(\sum_{m \geq 1} \frac{(-1)^m F_m(t)}{(m-1)!} \eta^{m-1} \right) \log \eta + \text{a function holomorphic near } \eta = 0.$$

This supplies exact continuation on the universal cover of $\eta \neq 0$. Since

$$1 - \rho = \frac{w - A(t)}{DA(t)} + O((w - A(t))^2),$$

(2.9) has residue $E(t)$ and logarithmic coefficient (2.6). Changing $\log(1 - \rho)$ to $\log(w - A(t))$ changes only a holomorphic remainder.

The other root summands have parameters $1 - \omega^k \rho$; their zeros all project to $w = A(t)$. At $\rho = 1$ they are locally holomorphic for $k \neq 0$. The compact part and the $D = 1$ correction are holomorphic at every nonzero diagonal. This proves the claimed exceptional set and uniformity. A change of logarithm branch adds a function holomorphic on the cover, and cannot alter the pole residue. $\square$

Remark 2.3 (Canonical local coefficient versus continued germ). The local logarithmic coefficient of the complete kernel is canonical at its nonzero diagonal. The continuation of its distinguished germ, however, is $\mathcal{B}(w, t, \rho)$ on $\mathcal{X}$ and need not descend to the base (w, t) -space. If another root summand becomes singular, the identity $\omega^{-k} \rho^{D-1} = (\omega^k \rho)^{D-1}$ gives the same canonical local coefficient after reselecting that summand. It does not identify different continued root sheets. Every integral below retains its root parameter. Appendix A gives an exact sign-changing example.

Lemma 2.4 (Normality near zeros of the top). *Near a zero of A , the original kernel integral is normally convergent for w on any fixed compact set away from zero. On every bounded t -disk it is holomorphic in t for all sufficiently large $|w|$.*

Proof. For $D > 1$, locally uniformly,

$$|P(t, r)/w|^{1/D} \leq |A(t)/w|^{1/D} r + C(1 + r)^{1-1/D}.$$

For $D = 1$ the second term is bounded. Near a zero of A , or for $|w|$ sufficiently large on a bounded t -disk, make the first coefficient less than $1/2$. The growth estimate from (2.7) is then integrably dominated by e^{-r} times the displayed sublinear exponential and a polynomial. To compare with a continued germ, first fix the entire finite continuation path in w away from zero, and shrink the t -neighborhood so this bound holds along that whole path. Uniqueness of continuation identifies that germ with the normally convergent integral. $\square$

3 Leaf identities and continuation with retained paths

Choose a regular value b different from $0, A(0), A(1)$. Let T be an embedded finite star centered at b , with one arm to each distinct member of

$$\{0, A(0), A(1)\} \cup \text{CritVal}(A).$$

The arms have disjoint interiors and are straight near their leaves. The inverse graph $\Gamma = A^{-1}(T)$ is a tree. Here is a useful direct justification. If T has s arms, then Γ has ds edges, d center vertices, and $ds - (d - 1)$ leaf vertices: the total finite ramification of a degree- d polynomial is $d - 1$. It is connected because the local monodromies about the leaf values generate a transitive covering group; their cycles join precisely the incident lifted stars. Thus its Euler characteristic is 1, and it is a tree. This is the usual polynomial inverse-tree construction [4, Section 2.1].

Let μ be its unique simple path from the actual point 0 to the actual point 1. Both μ and $[0, 1]$ lie in a bounded disk. Lemma 2.4 therefore permits replacing the interval by μ in (2.3) for large w . Orient each target arm from its leaf v toward b , and label the inverse branches over b by $1, \dots, d$. The path gives a vector $n_v \in \{0, 1, -1\}^d$, recording the signed use of the lift in each sheet. With these orientations,

$$\widehat{G}(w) = \sum_v \int_v^b \sum_{i=1}^d n_{v,i} \mathcal{C}(w, t_i(u)) t'_i(u) du. \quad (3.1)$$

This equality holds at infinity and on the chosen continuation off T . It retains every lift over each leaf.

Fix $v \neq 0$ and write $c_i = \lim_{u \rightarrow v} t_i(u)$ on its arm. For w near v , define the germ

$$\mathcal{L}_v(w) = \sum_i n_{v,i} \left\{ E(t_i(w)) t'_i(w) + \int_{c_i}^{t_i(w)} \mathcal{B}(w, t, \rho) dt \right\}. \quad (3.2)$$

Initially the integrals are local near c_i . Their lifts to $\mathcal{X}_w$ are selected by $\rho = 1$ at the upper endpoint. Since $A(c_i) = v \neq 0$, regular and ramified endpoints are both allowed.

Lemma 3.1 (Complete local singular extraction). *If $\widehat{G}$ is rational, then $\mathcal{L}_v \equiv 0$ for every nonzero leaf v .*

Proof. Put $z = w - v$ and $x = u - v$. Near the leaf, set

$$\begin{aligned} a(x) &= \sum_i n_{v,i} E(t_i(v+x)) t'_i(v+x), \\ b_0(z, x) &= \sum_i n_{v,i} \mathcal{B}(v+z, t_i(v+x), \rho) t'_i(v+x), \quad \rho^D = (v+x)/(v+z), \quad \rho \sim 1. \end{aligned}$$

These have convergent Puiseux expansions in x with a common finite denominator; b_0 is also holomorphic in z . Every x -exponent is greater than -1 , since a local inverse of degree k has derivative $O(x^{1/k-1})$.

All other edges in (3.1) are holomorphic in w near v on the chosen continuation. This follows from Lemma 2.2 where $A(t) \neq 0$, and from Lemma 2.4 near zero top values. The integrated local holomorphic remainder is holomorphic as well. After straightening the short arm, its relevant part is

$$\int_0^\epsilon \left(\frac{a(x)}{z-x} + b_0(z, x) \log(z-x) \right) dx.$$

A common nonzero linear rescaling of x, z does not affect the following vanishing argument. Put $q(z, x) = \int_0^x b_0(z, s) ds$. This tends to zero at the lower endpoint. Integration by parts gives a holomorphic boundary term plus

$$\int_0^\epsilon \frac{a(x) + q(z, x)}{z-x} dx. \quad (3.3)$$

In particular the sign of the correction is positive.

There are finitely many fractional classes. Choose their representatives $-1 < \delta \leq 0$ and write

$$a(x) + q(z, x) = \sum_{\delta} x^{\delta} h_{\delta}(z, x),$$

where each h_{δ} is holomorphic in the ordinary variables z, x . Divide $h_{\delta}(z, x) = h_{\delta}(z, z) + (x - z)k_{\delta}(z, x)$. The k_{δ} -term in (3.3) is holomorphic in z , by integrability of x^{δ} . The only nonholomorphic term from this class is therefore

$$h_{\delta}(z, z)I_{\delta}(z), \quad I_{\delta}(z) = \int_0^{\epsilon} \frac{x^{\delta}}{z - x} dx.$$

For $-1 < \delta < 0$, the singular part of I_{δ} is a nonzero multiple of z^{δ} on a fixed slit; scaling and the beta integral give a factor proportional to $\pi/\sin(\pi\delta)$. For $\delta = 0$ it is $\log z$, up to a holomorphic function. Distinct fractional classes cannot cancel, and none is a meromorphic germ. If (3.3), plus holomorphic terms, is rational, every $h_{\delta}(z, z)$ is zero. Hence

$$a(z) + \int_0^z b_0(z, x) dx = 0.$$

The change of variables $t = t_i(v + x)$ identifies this with (3.2). This proves vanishing of the whole corrected germ, not only of its leading coefficient. $\square$

Remark 3.2 (The triangular coefficient equations). If $a(x) = \sum a_{\alpha}x^{\alpha}$ and $b_0(z, x) = \sum b_{p,\beta}z^p x^{\beta}$, the preceding identity is equivalent to

$$a_{\alpha} + \sum_{p+\beta+1=\alpha} \frac{b_{p,\beta}}{\beta+1} = 0 \quad (\alpha > -1). \quad (3.4)$$

It is not an identity for a alone. For example, $-\sqrt{x}/(z-x) + (2\sqrt{x})^{-1} \log(z-x)$ is $\partial_x(\sqrt{x} \log(z-x))$, and its integral is holomorphic near $z = 0$, although both densities are nonzero.

Write G_A for the inverse-branch monodromy group. Fix a terminal path from b toward a nonzero critical value λ , with a final straight segment, avoiding zero and all critical values before its endpoint. For any negative Puiseux exponent α , define the linear functional

$$\Phi_{\lambda,\alpha}(e_i) = [(w - \lambda)^{\alpha}] \frac{\exp(A_1(t_i(w))/(Dw))}{A'(t_i(w))} \quad (3.5)$$

using continuation along that path and a common Puiseux parameter.

Theorem 3.3 (Transport of the full leaf identity). *If $\hat{G}$ is rational, $\Phi_{\lambda,\alpha}$ annihilates $\mathbb{Q}[G_A]n_v$ for every nonzero leaf v and every $\alpha < 0$.*

Proof. Let $U = \mathbb{C} \setminus (\{0\} \cup \text{CritVal}(A))$. Continue the zero germ $\mathcal{L}_v$ along its arm to b , and then along any path in U . Inverse branches continue and stay in V . For an individual integral in (3.2), retain its path from the fixed c_i and append the path of its moving upper endpoint. Lift the accumulated path to $\mathcal{X}_w$ with $\rho = 1$ at that upper endpoint. The fixed lower root is allowed to change after a loop; it remains part of the data.

Here is a precise local description. On a small w -disk W away from zero choose a holomorphic root $s(w)^D = w$, and put

$$\Sigma = \{(t, \sigma) : t \in V, \sigma^D = A(t)\}, \quad \rho = \sigma/s(w).$$

Then $\mathcal{X}|_W$ is identified with $W \times \Sigma$. The curve Σ is locally parametrized by t , since $\sigma \neq 0$, even at a critical point of A . The differential $\mathcal{B}(w, t, \sigma/s(w)) dt$ is holomorphic on this product.

A fixed compact lifted path integrates to a holomorphic function of w ; a short appended piece has a jointly holomorphic local primitive. This constructs the continuation along each compact portion of the chosen path. At the moving endpoint $\sigma = s(w)$, so $\rho = 1$ is preserved. Different roots at a fixed lower endpoint are never reset.

Now approach $\lambda \neq 0$, and suppose an endpoint tends to c with $A(c) = \lambda$. Split the accumulated history into a fixed compact lifted path ending near c , and a short terminal piece. The first part is holomorphic on a disk about λ by the product description. A point with $A(t) = w$ on that history causes no pole in $\mathcal{B}$. Near (λ, c) the root with $\rho \sim 1$ is holomorphic. A primitive $J(w, t)$ of the corresponding differential is jointly holomorphic, and the terminal piece is

$$J(w, t_i(w)) - J(w, t_*).$$

If A has local degree k at c , then $t_i(w) = c + O((w - \lambda)^{1/k})$. Thus this primitive has only nonnegative Puiseux exponents. The same holds for every retained history. Every negative coefficient of the continued identity comes solely from $E(t_i(w))t'_i(w)$.

For each $g \in G_A$, choose a loop at b realizing g , and then follow the fixed terminal path. The continued $\mathcal{L}_v$ is still zero, so its negative coefficient is $\Phi_{\lambda, \alpha}(gn_v) = 0$. The complete histories can depend on more than the permutation g ; no action of G_A on those histories is asserted or needed. Their negative coefficients vanish, while the pole contribution depends only on g . Linear extension proves the theorem. Removing the extra regular point zero from the base, when zero is not critical, does not remove any permutation from its monodromy image. $\square$

4 The path module and the simultaneous trace identity

We use the bilinear pairing $\langle x, y \rangle = \sum_i x_i y_i$. Vector spaces are over $\mathbb{Q}$ until explicitly complexified. For the actual signed vectors in (3.1), put

$$M = \sum_v \mathbb{Q}[G_A]n_v, \quad M_{\neq 0} = \sum_{v \neq 0} \mathbb{Q}[G_A]n_v.$$

Lemma 4.1 (The zero leaf is algebraically redundant). *One has $\sum_v n_v = 0$ coordinatewise, and $M_{\neq 0} = M$.*

Proof. Each visited center vertex of μ has one entering and one leaving edge, contributing opposite signs to its sheet coordinate. Neither actual endpoint lies over b . Thus $n_0 = -\sum_{v \neq 0} n_v$; apply all monodromy elements. This makes no analytic assertion at $w = 0$. $\square$

Label sheets so a full infinity cycle is $(1 \ 2 \ \cdots \ d)$. For $k \mid d$, let $V_k \subset \mathbb{Q}^d$ consist of k -periodic vectors, and set

$$D(G_A) = \{k \mid d : V_k \text{ is } G_A\text{-invariant}\}, \quad U_d = \bigcap_{\substack{k \in D(G_A) \\ k < d}} V_k^\perp, \quad U_d^\mathbb{C} = U_d \otimes_{\mathbb{Q}} \mathbb{C}. \quad (4.1)$$

The indicators of residue classes modulo k span V_k . Because the cyclic subgroup is regular, every block system is of this form: the block containing the label 0 in $\mathbb{Z}/d\mathbb{Z}$ is a subgroup, and its translates are all blocks. Invariance of its indicator span is equivalent to invariance of the partition. In polynomial monodromy these blocks correspond to decompositions $A = P \circ R$. The intermediate-field correspondence and Lüroth's theorem give a rational generator; the unique point over infinity permits choosing that generator R with its only pole at infinity. Then R , and also P , are polynomials. For V_k one has $\deg P = k$ and $\deg R = d/k$.

Theorem 4.2 (Rational cyclic-module input, Pakovich–Muzychuk). *For a permutation group $G \leq S_d$ containing a full d -cycle, the space U_d defined as in (4.1) is an irreducible rational G -module.*

This is the top-component case of [4, Theorem 3.1]. Its proof uses the classification of rational Schur rings over a cyclic group. That representation-theoretic theorem, not the whole polynomial moment problem, is retained as an external input. The next lemmas prove its application to the actual path, following the geometry of [4, Section 2.2 and Proposition 4.1].

Lemma 4.3 (A rational invariant subspace dichotomy). *If $N \subset \mathbb{Q}^d$ is G_A -invariant, then either $U_d \subset N$ or $N \perp U_d$.*

Proof. The orthogonal projectors P_N and P_U have rational entries and commute with G_A . They therefore commute with the full cycle. Its centralizer consists of circulant matrices and is commutative, so $P_N P_U = P_U P_N$. The restriction of P_N to U_d is a G_A -equivariant idempotent. By Theorem 4.2 it is either zero or the identity. These are the two assertions. No absolute irreducibility is used. $\square$

For an actual endpoint $p \in \{0, 1\}$, let I_p be the set of sheet labels whose inverse branches on its arm tend to p , and let $e_p = |I_p|$ be its local degree. Write

$$m_p = \frac{1}{e_p} \sum_{i \in I_p} e_i \in \mathbb{Q}^d.$$

Lemma 4.4 (Endpoint averaging). *If $A(0) = A(1)$, then $m_0 - m_1 \in M$. If $A(0) \neq A(1)$, then $m_0, m_1 \in M$.*

Proof. Average the vector n_v over the finite cyclic group generated by the local monodromy at its leaf. On the labels tending to one point above v , the averaging replaces every coordinate by the mean over that local cycle. At an interior visit of μ , one edge enters and another leaves the point; their sum on that cycle is zero. At an actual endpoint the sum is $+1$ for 0 , and -1 for 1 , under our orientations. Thus if the endpoint values agree, the average is $m_0 - m_1$. Otherwise the two separate averages are m_0 and $-m_1$. All averages belong to M . $\square$

Lemma 4.5 (Planar separation of endpoint labels). *Let $\zeta = e^{2\pi i/d}$, and identify a label i with ζ^{i-1} . If $A(0) = A(1)$, the two endpoint label sets lie in disjoint open arcs of the circle. If their values differ, the sets have this property or share one point and, after that point is removed, lie on opposite open arcs cut out by it and one further circle point. In particular, their convex hulls are disjoint in the first case and meet at most in that single circle point in the second.*

Proof. Append to T a simple arc from b to infinity, meeting T only at b . Its lifts are disjoint rays from the d center vertices to infinity. Cut them off on a sufficiently large circle. Their cyclic order is that of the full infinity cycle, up to reversing it, which does not affect the assertion. For an endpoint p , take the union K_p of its incident arm lifts and the attached rays. It is an embedded star joining p to exactly the boundary labels I_p ; intermediate center vertices have degree two.

If the endpoints have the same value, K_0 and K_1 are disjoint. Between two consecutive rays of K_0 , the two rays and the intervening boundary arc enclose a sector by the Jordan curve theorem. The connected star K_1 , disjoint from K_0 , lies in a single such sector. Its boundary labels consequently occur in one gap between consecutive labels of I_0 . A chord with endpoints in the adjacent gaps strictly separates the two finite label sets. The singleton case is immediate by taking a small arc about that single label.

For different endpoint values, two stars can share a center only if one lifted star contains both endpoints. There cannot be two such centers: the two paths from 0 through these centers to 1 would form a cycle in $A^{-1}(T)$. If there is no common center, the preceding argument applies. Otherwise $K_0 \cap K_1$ is the one common ray from that center to the boundary. The same Jordan separation, now cutting along the common ray, puts all other labels of K_1 in a

single sector of K_0 adjacent to that ray. The two sets are separated by a chord through their common boundary label and a point in the opposite gap. The two convex hulls can meet only at the common label. This also covers the case when one star has only the common ray. $\square$

Proposition 4.6 (The actual path contains the top component). *For $d \geq 2$, $U_d \subset M$.*

Proof. Put $z = (1, \zeta, \dots, \zeta^{d-1})$. The sum of its entries on every residue class modulo a proper divisor k is zero. Thus $z \in U_d^{\mathbb{C}}$. The number $\langle m_p, z \rangle$ is the arithmetic mean of the roots of unity labeled by I_p . If the endpoint values agree, Lemma 4.5 gives disjoint convex hulls, so these two means differ. Lemma 4.4 therefore supplies a vector in M pairing nontrivially with z . If the values differ, both means cannot be zero: their convex hulls are disjoint or meet only on the unit circle. At least one of the two vectors in Lemma 4.4 again pairs nontrivially with z . Hence M is not orthogonal to U_d , and Lemma 4.3 gives $U_d \subset M$. $\square$

This is the needed path-module statement of [4, Proposition 4.1], with its endpoint and sign conventions established here. It allows ramified endpoints and equal or zero endpoint values. A regular extra leaf at zero contributes no active lift unless it is an endpoint value.

Corollary 4.7. *If $\widehat{G}$ is rational, every functional (3.5) with $\lambda \neq 0$ and $\alpha < 0$ annihilates $U_d^{\mathbb{C}}$.*

Proof. Theorem 3.3 gives annihilation of $M_{\neq 0}$. Lemma 4.1 and Proposition 4.6 give $U_d \subset M = M_{\neq 0}$. Extend scalars. $\square$

Lemma 4.8 (Simultaneous trace orthogonality). *For every regular w , in the continued sheet ordering,*

$$u(w) = \left(\frac{1}{A'(t_1(w))}, \dots, \frac{1}{A'(t_d(w))} \right) \in U_d^{\mathbb{C}}. \quad (4.2)$$

Every Puiseux coefficient vector of u also belongs to $U_d^{\mathbb{C}}$.

Proof. Take any proper $k \in D(G_A)$ and the corresponding decomposition $A = P \circ R$, with $r = \deg R = d/k \geq 2$. A block over w is a fiber $R^{-1}(\xi)$ with $P(\xi) = w$. Regularity gives $P'(\xi) \neq 0$ and r distinct points in that fiber. The block trace is

$$\sum_{i \text{ in block}} \frac{1}{A'(t_i(w))} = \frac{1}{P'(\xi)} \sum_{R(c)=\xi} \frac{1}{R'(c)} = 0. \quad (4.3)$$

Indeed, in the partial fractions

$$\frac{1}{R(t) - \xi} = \sum_{R(c)=\xi} \frac{1}{R'(c)(t - c)},$$

the coefficient of t^{-1} is zero since the left side is $O(t^{-r})$. Block indicators span V_k , so $u \perp V_k$. Intersect over all proper k . The relations are constant rational equations, and consequently hold coefficientwise in every continued Puiseux expansion. $\square$

The intersection in this argument treats all proper decompositions at once. It imposes no constancy condition on a weighted vector along any one chosen block system.

5 The arithmetic obstruction and the nonflat conclusion

The next proposition applies to the top two layers of any algebraically specialized polynomial; it does not impose genericity on their coefficients.

Proposition 5.1 (Nonzero critical values are detected). *Suppose $A, A_1 \in \overline{\mathbb{Q}}[t]$, $\deg A = d \geq 2$, and all negative coefficient functionals in (3.5) at nonzero critical values annihilate U_d . Then A has no nonzero critical value.*

Proof. Suppose $\lambda \neq 0$ is a critical value. Let $K \geq 2$ be the largest local degree of A among the points above λ , and put $\alpha = 1/K - 1 < 0$. At a point c of local degree K , write

$$A(c + s) = \lambda + a_c s^K + O(s^{K+1}), \quad a_c \neq 0.$$

The K inverse branches there have convergent expansions

$$t_i(w) = c + b_c \zeta_i (w - \lambda)^{1/K} + O((w - \lambda)^{2/K}), \quad b_c^K = a_c^{-1}, \quad \zeta_i^K = 1.$$

Let β be the coefficient vector at exponent α in $u(w)$ from (4.2). On these branches, $\beta_i = b_c \zeta_i / K$. At a point of smaller local degree $k < K$, the first exponent $1/k - 1$ is strictly larger than α , so $\beta_i = 0$. In particular, $\beta \neq 0$, its entries are algebraic, and Lemma 4.8 gives $\beta \in U_d^{\mathbb{C}}$.

Since $A(t_i(w)) = w$ and $\lambda \neq 0$, the exponential weight is holomorphic in the local inverse coordinate. Its constant term is $\exp(A_1(c_i)/(D\lambda))$, and all other terms have positive Puiseux exponent. Thus the coefficient at α in the weighted inverse-derivative vector is exactly

$$\phi_i = \begin{cases} \beta_i \exp(A_1(c_i)/(D\lambda)), & \beta_i \neq 0, \\ 0, & \beta_i = 0. \end{cases} \quad (5.1)$$

The maximal choice of K is what excludes contributions from smaller ramification indices or nonconstant terms of the weight.

By hypothesis ϕ is bilinearly orthogonal to $U_d^{\mathbb{C}}$. Since U_d is defined over $\mathbb{Q}$, its complexification is preserved by conjugation. Consequently $\bar{\beta} \in U_d^{\mathbb{C}}$, and

$$0 = \langle \phi, \bar{\beta} \rangle = \sum_{\beta_i \neq 0} |\beta_i|^2 \exp\left(\frac{A_1(c_i)}{D\lambda}\right). \quad (5.2)$$

All exponents and coefficients in this expression are algebraic. After equal exponents are grouped, each coefficient is a strictly positive real algebraic number. At least one group occurs. Lindemann–Weierstrass [1] makes the exponentials of distinct algebraic numbers linearly independent over $\overline{\mathbb{Q}}$, contradicting (5.2). $\square$

Remark 5.2. Grouping by critical points instead gives

$$\sum_{\substack{A(c)=\lambda \\ e_c=K}} \frac{(a_c \bar{a}_c)^{-1/K}}{K} \exp\left(\frac{A_1(c)}{D\lambda}\right) = 0,$$

where the positive real root is intended. This records explicitly that local root choices do not affect the positive coefficients. The positivity is created by the conjugate pairing, not by the original oriented interval or by the complex exponentials.

Lemma 5.3 (Only zero critical values). *If all critical values of a polynomial A of degree $d \geq 2$ are zero, then $A(t) = a(t - c)^d$ for some $a \neq 0$.*

Proof. Every root of A' is a root of A . If the distinct roots of A have multiplicities $m_1, \dots, m_q$, their multiplicities in A' are exactly $m_j - 1$. They therefore account for $d - q$ roots of A' . By hypothesis they account for all $d - 1$, so $q = 1$. $\square$

Lemma 5.4 (The endpoints exclude pure powers). *Let $f \in \overline{\mathbb{Q}}[x, y]$ have $A(t) = a(t - c)^d$, $a \neq 0$, $d \geq 1$. Then $\hat{G}$ is not rational.*

Proof. Every nonzero value of A is regular. At an actual endpoint p with $A(p) = v \neq 0$, the constant logarithmic coefficient of the leaf integral is, with its orientation, $E(p)/A'(p)$. This follows either from Lemma 3.1 or directly from (2.4): the pole density has that constant term, while the Volterra correction starts at order $w - v$. Unramified preimages of a leaf have graph degree one, so only the actual endpoints can be active above this regular value.

If $A(0) \neq A(1)$, one endpoint has a nonzero value different from the other endpoint value. Its nonzero logarithmic coefficient cannot cancel. If the endpoint values agree, their common value v cannot be zero, since A has just one distinct root. Rationality would give

$$\frac{\exp(A_1(0)/(Dv))}{A'(0)} = \frac{\exp(A_1(1)/(Dv))}{A'(1)}.$$

The exponential of the algebraic difference of the two exponents is therefore algebraic. Hermite–Lindemann forces that difference to be zero; the equality then forces $A'(0) = A'(1)$. But

$$A'(0) = \frac{dv}{-c}, \quad A'(1) = \frac{dv}{1-c},$$

which are unequal. When $d = 1$, equal endpoint values are already impossible. These cases exhaust the possibilities. $\square$

Proof of Theorem 1.1. Suppose all positive factorial moments of a nonflat f vanish. Lemma 2.1 gives such a polynomial with algebraic coefficients. Its transform is $\widehat{G} = 1/w$. If $d \geq 2$, Corollary 4.7 and Proposition 5.1 exclude every nonzero critical value. Lemma 5.3 then makes A a pure power. If $d = 1$, it is already a translated pure power. Lemma 5.4 contradicts rationality in both cases.

Finally, if only finitely many moments of a nonflat f were nonzero, choose $N \geq 1$ such that $\mathcal{L}(f^n) = 0$ for every $n \geq N$. The nonzero polynomial f^N has nonconstant projective top A^N and every positive moment zero, contrary to the first assertion. $\square$

6 The flat case and the EIT input

The radial estimate below is independent of the monodromy argument. It shows exactly where the EIT theorem enters.

Lemma 6.1 (Uniform radial limit). *On every compact t -set on which A does not vanish,*

$$\frac{1}{\Gamma(Dn+2)} \int_0^\infty e^{-r} r P(t, r)^n dr = A(t)^n \left[\exp\left(\frac{A_1(t)}{DA(t)}\right) + o(1) \right], \quad (6.1)$$

with a uniform error inside the brackets.

Proof. For $\mathbf{k} = (k_1, \dots, k_D) \in \mathbb{N}^D$, put $K = \sum_j k_j$ and $J = \sum_j j k_j$. The multinomial expansion and termwise radial integration give terms

$$A^{n-K} \prod_{j=1}^D \frac{A_j^{k_j}}{k_j!} C_{n,\mathbf{k}}, \quad C_{n,\mathbf{k}} = \frac{n!}{(n-K)!} \frac{\Gamma(Dn-J+2)}{\Gamma(Dn+2)} \quad (K \leq n).$$

Set $C_{n,\mathbf{k}} = 0$ for $K > n$. The inequalities $K \leq J \leq DK$ show $0 \leq C_{n,\mathbf{k}} \leq 1$. Indeed, the smallest K factors in the denominator product dominate the numerator factors:

$$Dn - J + 2 + j \geq D(n - K) + 2 + j \geq n - K + 1 + j \quad (0 \leq j < K).$$

These denominator factors are present because $J \geq K$; all remaining ones are at least one. The empty-product case $K = 0$ is immediate. For fixed $\mathbf{k}$,

$$C_{n,\mathbf{k}} \sim D^{-J} n^{K-J}.$$

After division by A^n , only the terms $k_2 = \dots = k_D = 0$ survive, and their sum is $\exp(A_1/(DA))$. On the compact set, all absolute terms are bounded by

$$\prod_{j=1}^D \frac{(M_j/a_*)^{k_j}}{k_j!}, \quad M_j = \max |A_j|, \quad a_* = \min |A| > 0.$$

This is summable over $\mathbb{N}^D$ and proves uniform dominated convergence, including $D = 1$. $\square$

Proof of Corollary 1.3. The nonflat case is Theorem 1.1, and a nonzero constant polynomial is immediate. Suppose $D \geq 1$ and A is constant. It is nonzero, so scaling f permits $A = 1$. If all positive moments vanish, specialize algebraically with this normalization retained. Equation (6.1) gives

$$\lim_{n \rightarrow \infty} M_n = \int_0^1 \exp\left(\frac{A_1(t)}{D}\right) dt.$$

When A_1 is constant this limit is a nonzero exponential. Otherwise Theorem 1.2 applies with $q = A_1$, $a = 0$, $b = 1$ and $\xi = 1/D$. Its hypotheses hold over $\overline{\mathbb{Q}}$, and the amplitude is exactly dt . The limit is again nonzero, contradicting $M_n = 0$ for every $n \geq 1$. The infinitely-many assertion follows by applying the result to f^N if a tail of its factorial moments were zero. $\square$

Research provenance and assistance. During the development of this argument, the `openai/math` release suggested examining local jet separation, differential-system multiplicity, and continuation of complete branch data [9, 8, 10]. These were prompts for proof exploration, not theorems transferred into the FC(2) argument. In particular, local étale rigidity does not imply injectivity of a sum over sheets, and the cited billiard continuation results are not applied here. The main proof uses none of the release’s results. Appendix B records the separate finite-jet application of Moura’s classical estimate, together with its formal proof in the release. ChatGPT assisted with exploration, drafting, symbolic checking and review; the author is responsible for the mathematical claims and their attribution.

A Exact continuation examples

The examples in this appendix are independent checks of the continuation conventions. None is an input to Sections 5–6.

A.1 The pole alone has the wrong jump

For $f = 3x + 2y + 1$, one has $D = 1$ and $A(t) = 2 + t$. Its exponential generating series is

$$\sum_{n \geq 0} \frac{\mathcal{L}(f^n)}{n!} z^n = \frac{e^z}{(1-3z)(1-2z)}.$$

Integrating the series and substituting $z = 1/w$ gives the exact identities

$$\widehat{G}(w) = \int_0^{1/w} \frac{e^z}{(1-3z)(1-2z)} dz, \quad \widehat{G}'(w) = -\frac{e^{1/w}}{(w-3)(w-2)}.$$

Counterclockwise continuation around 3 therefore gives $\Delta_3 \widehat{G} = -2\pi i e^{1/3}$. In contrast, the pole integral alone is

$$C_0(w) = \int_2^3 \frac{e^{1/u}}{w-u} du, \quad \Delta_3 C_0 = -2\pi i e^{1/w}.$$

The difference is a nonconstant analytic germ. For this family, $\mathcal{B}(w, t) = e^{1/A(t)}/A(t)^2$, and the corrected branch germ at a lower endpoint with value v is

$$e^{1/w} + \int_v^w \frac{e^{1/u}}{u^2} du = e^{1/v}.$$

Thus the correction, while one local singular order softer, changes the entire jump. A leading-pole identity cannot be substituted for (3.2).

A.2 A coefficient that changes root sheets

Take $D = 2$ and $P = Ar^2 + c$, where $c \neq 0$ and $A = A(t) \neq 0$. The coefficient formula can be computed independently of radial reversion. On writing $n = m + k$, the duplication formula gives

$$\binom{m+k}{k} \frac{\Gamma(2m+2)}{\Gamma(2m+2k+2)} = \frac{4^{-k}\Gamma(m+3/2)}{k!\Gamma(m+k+3/2)}.$$

For $k \geq 1$, use the beta integral for the ratio and sum the geometric series over m . For large $|w|$ this gives

$$\mathcal{C}(w, t) = \frac{1}{w-A} + \int_0^1 \frac{u^{1/2}}{w-Au} \sum_{k \geq 1} \frac{(c/(4w))^k (1-u)^{k-1}}{k!(k-1)!} du. \quad (\text{A.1})$$

All exchanges converge normally there. At the endpoint $u = 1$, the logarithmic coefficient is minus $1/A$ times the analytic numerator evaluated at $u = w/A$. Reindexing $k = j + 1$ yields

$$\mathcal{B}(w, t, \rho) = -\frac{c\rho}{4A^2} \sum_{j \geq 0} \frac{[c(\rho^2 - 1)/(4A)]^j}{j!(j+1)!}, \quad \rho^2 = A/w. \quad (\text{A.2})$$

This is an all-orders convergent identity. In particular, $\mathcal{B}(A, t, 1) = -c/(4A^2)$, agreeing with (2.6).

The right side of (A.2) is odd in ρ . A fixed- t loop around zero changes the sign of the continued distinguished coefficient. When the loop returns to a neighborhood of the diagonal, the canonical coefficient of the complete kernel is unchanged because the singular root-filter summand has been reselected. This distinguishes the two objects in Remark 2.3 concretely.

A.3 A moving endpoint with a changed lower root

Let

$$P = (L(t)r + c)^D, \quad L(t) = at + b, \quad ac \neq 0, \quad D \geq 2.$$

This is the radial polynomial of $[ax + b(x+y) + c]^D$. Fix a lower endpoint p with $L(p) \neq 0$, and write $s = L(t_i(w))$, so $s^D = w$. On the lifted integration path,

$$\Phi = r + c/L, \quad \chi = Y - c/L, \quad F = e^{c/L}(1 - c/(LY)), \quad \rho = L/s.$$

Formula (2.6) now reads

$$\mathcal{B} dt = -\frac{d_t(e^{c/L(t)})}{Da s^{D-1}}.$$

Since $A'(t_i) = Da s^{D-1}$, the full corrected expression is

$$E(t_i(w))t'_i(w) + \int_p^{t_i(w)} \mathcal{B} dt = \frac{e^{c/L(p)}}{Da s^{D-1}}. \quad (\text{A.3})$$

For a positive loop around zero, s changes to ωs , where $\omega = e^{2\pi i/D}$. The lower root parameter changes to $\omega^{-1}\rho$, and (A.3) changes by the factor ω . Resetting the lower root, or deleting the integral, would both give the wrong continuation. The general transport proof retains exactly these data.

B Finite certificates for branch densities

This appendix concerns auxiliary differential systems, not a bound for the first nonzero factorial moment. The multiplicity estimate is due to Moura [6, 7]; the `openai/math` release gives a formal power-series proof and a Lean statement [8]. We record the ordinary-point form and its application to the weighted inverse derivatives. The proof below is included to specify the exact hypotheses and certificate.

B.1 The classical multiplicity estimate

Theorem B.1 (Moura’s estimate in a polynomial presentation). *Let F have characteristic zero, $m, n \geq 1$, $q \in F[z]$ with $q(0) \neq 0$ and $\deg q \leq n$, and $P \in M_m(F[z])$ with $\deg P_{ij} \leq n - 1$. Suppose*

$$G \in \mathrm{GL}_m(F[[z]]), \quad qG' = PG.$$

For a constant row u and constant column v , a nonzero series $y = uGv$ satisfies

$$\mathrm{ord}_0 y \leq (n - 1) \binom{m}{2} + m - 1. \quad (\text{B.1})$$

Proof. Put $h = uG$, and let V be the F -span of its component series. Its dimension k is between 1 and m . Gaussian elimination on successive coefficients gives a basis $b_1, \dots, b_k$ of V with strictly increasing orders $a_1 < \dots < a_k$. The filtration by order is eventually zero: in a finite-dimensional space any stabilizing nonzero tail would lie in every power of z , which is impossible. A nonzero element of V has order at most a_k .

The Wronskian $W = \det(b_\ell^{(j)})_{0 \leq j < k, 1 \leq \ell \leq k}$ has order

$$\mathrm{ord}_0 W = \sum_{i=1}^k a_i - \binom{k}{2}. \quad (\text{B.2})$$

Indeed, its leading coefficient after the powers of z are factored is the product of the leading coefficients of the b_i times $\det((a_i)_j)$, a nonzero Vandermonde determinant in characteristic zero. Changing the basis of V multiplies W by a nonzero constant.

Define rows $r_0 = u$, $r_{j+1} = r'_j + r_j P/q$. Then $h^{(j)} = r_j G$. Writing $r_j = Z_j/q^j$ gives

$$Z_{j+1} = qZ'_j - jq'Z_j + Z_j P, \quad \deg Z_j \leq j(n - 1).$$

Let R_* have rows $r_0, \dots, r_{k-1}$ and put $S = \binom{k}{2}$. Some k components of h form a basis of V , so $R_* G$ has a nonzero k -minor by (B.2). Thus R_* also has such a minor. Each k -minor of R_* has denominator q^S and a polynomial numerator of degree at most $(n - 1)S$. Since q is a unit at zero, the minimum order δ of its nonzero k -minors obeys $0 \leq \delta \leq (n - 1)S$.

Multiplication by G preserves the ideal of k -minors in $F[[z]]$: apply Cauchy–Binet first with G and then with G^{-1} . Hence the minimum order for the minors of $R_* G$ is also δ . A nonzero minor of that order is a Wronskian of k components of h ; those components form a basis of V . Equation (B.2) therefore gives

$$\delta = \sum_{i=1}^k (a_i - (i - 1)), \quad \mathrm{ord}_0 y \leq a_k \leq \delta + k - 1 \leq (n - 1) \binom{k}{2} + k - 1.$$

Each summand in the first expression is nonnegative. Finally $k \leq m$ proves (B.1). □

The argument is the derivative-row and Wronskian method appearing in Moura’s work and in the release’s formal proof. The corresponding Lean declaration is `OAI.NCHitting.system_multiplicity`. Reading that declaration is not a claim that the present FC(2) paper, or this application, has been formally verified.

B.2 An explicit rational connection

Fix $A, B \in \mathbb{C}[t]$, $\deg A = d \geq 1$, and an integer $D \geq 1$. On a disk avoiding zero and the critical values of A , put

$$h_i(w) = \frac{\exp(B(t_i(w))/(Dw))}{A'(t_i(w))}, \quad \Psi_{ji}(w) = t_i(w)^j h_i(w) \quad (0 \leq j < d).$$

The determinant of Ψ is a Vandermonde determinant times nonzero column factors, so it is invertible. In $\mathbb{C}(w)[t]/(A(t) - w)$, reduce

$$\frac{jt^{j-1}}{A'} - \frac{t^j A''}{(A')^2} + \frac{t^j B'}{DwA'} - \frac{t^j B}{Dw^2} = \sum_{k=0}^{d-1} R_{jk}(w)t^k. \quad (\text{B.3})$$

The first term is understood as zero for $j = 0$. Differentiating $t'_i = 1/A'(t_i)$ and h_i proves the exact system $\Psi' = R\Psi$.

Let $\Delta(w) = \text{Res}_t(A(t) - w, A'(t))$. It is a nonzero polynomial. The resultant identity provides a representative of $1/A'$ whose only parameter denominator is Δ . Reduction modulo $A - w$ introduces only constant leading-coefficient denominators. Consequently

$$q(w) = Dw^2 \Delta(w)^2, \quad P(w) = q(w)R(w)$$

has polynomial entries. Define

$$n = \max\{1, \deg q, 1 + \max_{P_{ij} \neq 0} \deg P_{ij}\}, \quad J = (n-1) \binom{d}{2} + d - 1,$$

omitting the last entry of the maximum when $P = 0$.

Corollary B.2 (Finite test for an identically zero density). *For a constant coefficient vector c , an ordinary point $w_0 \neq 0$, and $h(w) = \sum_i c_i h_i(w)$ on its disk, one has*

$$h \equiv 0 \iff h^{(j)}(w_0) = 0 \quad (0 \leq j \leq J).$$

Proof. Translate w_0 to zero and normalize the fundamental matrix to $G(w) = \Psi(w)\Psi(w_0)^{-1}$. Then $h = e_0^\top G(\Psi(w_0)c)$ is a constant scalar output. The denominator q is nonzero at w_0 . Apply Theorem B.1 contrapositively. $\square$

For algebraic A, B, w_0, c , the derivatives are finite algebraic linear combinations of exponentials of algebraic numbers. They can be tested exactly by grouping equal exponents and using Lindemann–Weierstrass. This certifies a particular density identity. It neither makes every sheet sum nonzero nor bounds the first nonzero moment of an arbitrary factorial polynomial.

For example, $A = t^2 + 1$, $B = t$ and $D = 2$ give

$$R = \begin{pmatrix} -\frac{1}{2(w-1)} & \frac{2-w}{4w^2(w-1)} \\ \frac{2-w}{4w^2} & 0 \end{pmatrix}.$$

One may take $q = 4w^2(w-1)$, $n = 3$ and $J = 3$. The first two observation rows $e_0^\top$ and $e_0^\top R$ are independent over $\mathbb{C}(w)$. By contrast, for $A = t(1-t)$ and constant $B = c$,

$$\frac{e^{c/(2w)}}{A'(t_+(w))} + \frac{e^{c/(2w)}}{A'(t_-(w))} = 0, \quad t_\pm(w) = \frac{1 \pm \sqrt{1-4w}}{2}.$$

Both summands are nonzero. All jets of their sum vanish, even on the étale locus. The actual interval uses opposite orientations on these two branches; this is why the signed path data in the main proof cannot be replaced by local jet rigidity.

B.3 A different, conditional certificate for a complete system

If an exact full transform is embedded in a rational system $Y' = MY$ and $h = \ell_0 Y$, define observation rows $\ell_{j+1} = \ell'_j + \ell_j M$. Then $h^{(j)} = \ell_j Y$. If a coordinate row $e_r^\top$ lies in the rational span of these rows and $Y_r \neq 0$, one necessarily has $h \neq 0$. Indeed, $h \equiv 0$ would force $Y_r \equiv 0$ by the displayed row relation. This criterion requires actual observation of a nonzero coordinate; finiteness of the system alone is not sufficient.

For a nonconstant linear polynomial $f = ax + by + c$, let $q = (w - a)(w - b)$ and take

$$Y = (\widehat{G}, e^{c/w}, 1)^\top, \quad M = \begin{pmatrix} 0 & -1/q & 0 \\ 0 & -c/w^2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \ell_0 = (1, 0, -1/w).$$

The system follows by integrating the exponential generating series, and $h = \widehat{G} - 1/w$ is the exact positive-moment output. Direct differentiation gives

$$\det \begin{pmatrix} \ell_0 \\ \ell_1 \\ \ell_2 \end{pmatrix} = \frac{-(a + b + c)w^2 + (2ab + c(a + b))w - cab}{q^2 w^4}.$$

Its numerator is identically zero only if $a = b = c = 0$: from $c = -a - b$, the middle coefficient gives $a^2 + b^2 = 0$, and the last gives $ab(a + b) = 0$, forcing $a = b = 0$. Thus in the nonconstant case all coordinates, including 1, are observed and h cannot vanish identically. No analogous universal observation assertion in higher degree is needed or claimed here.

C A collision with incomparable decompositions

Here is a concrete instance of Proposition 5.1. Let $D \geq 6$ and suppose

$$A(t) = T_6(2t - 1) - 1, \quad T_6(z) = 32z^6 - 48z^4 + 18z^2 - 1.$$

It admits both $T_6 = T_2 \circ T_3$ and $T_6 = T_3 \circ T_2$, where $T_2(z) = 2z^2 - 1$ and $T_3(z) = 4z^3 - 3z$. The corresponding block sizes are incomparable, and $A(0) = A(1) = 0$. The three points over the nonzero critical value -2 are

$$c_- = \frac{2 - \sqrt{3}}{4}, \quad c_0 = \frac{1}{2}, \quad c_+ = \frac{2 + \sqrt{3}}{4}.$$

They have local degree two, with quadratic coefficients $A''(c)/2$ equal to 288, 72, 288, respectively. The positive critical-point weights in (5.2) are therefore $1/576, 1/144, 1/576$. Rationality of $\widehat{G}$ would force

$$e^{-A_1(c_-)/(2D)} + 4e^{-A_1(1/2)/(2D)} + e^{-A_1(c_+)/(2D)} = 0.$$

For algebraic layers this contradicts Lindemann–Weierstrass after equal exponents are grouped. There are no restrictions on the subsequent layers. Algebraic specialization gives the corresponding nonvanishing statement with arbitrary complex coefficients. The computation illustrates why neither colliding critical points, incomparable block systems, nor zero endpoint values are omitted by the proof.

Reproducibility. The source bundle includes an exact $\mathbb{Q}(\sqrt{3})$ calculation of the identities in this appendix, and symbolic checks of the rational connection, observation determinant, root filter, radial normalization, and branch-change examples. These finite checks are separate from the analytic and representation-theoretic proofs.

References

- [1] A. Baker, *Transcendental Number Theory*, Cambridge Mathematical Library, Cambridge University Press, 1990, Chapter 1.
- [2] A. van den Essen, D. Wright and W. Zhao, *On the Image Conjecture*, J. Algebra **340** (2011), 211–224. doi:10.1016/j.jalgebra.2011.04.036. arXiv:1008.3962v2.
- [3] D. Liu and X. Sun, *The factorial conjecture and images of locally nilpotent derivations*, Bull. Aust. Math. Soc. **101** (2020), 71–79. doi:10.1017/S0004972719000546.
- [4] F. Pakovich and M. Muzychuk, *Solution of the polynomial moment problem*, Proc. Lond. Math. Soc. (3) **99** (2009), 633–657. doi:10.1112/plms/pdp010. arXiv:0710.4085v2. Theorem 3.1 is the rational representation input; the path arguments are in Sections 2.1–2.2 and Proposition 4.1.
- [5] C. D. Long and A.-A. Le Blanc, *Algebraic Exponential Integrals: Transcendence and Linear Relations*, EIT I, research manuscript, 2026, Corollary `cor:single`. Pinned LaTeX source, commit `22d98c4f27e606b5b25fa673c33f9c56c53cc86b`, consulted 7 October 2026. This is the external input to Theorem 1.2; its proof is not reproduced here.
- [6] C. Moura, *Bounds for the vanishing order for solutions of a linear differential system*, J. Dynam. Control Systems **9** (2003), no. 1, 73–88. doi:10.1023/A:1022155217548.
- [7] C. Moura, *On the multiplicity of the hyperelliptic integrals*, preprint, 2003, Section 2, Theorem 1. arXiv:math/0312323. Used for the stated ordinary-point multiplicity estimate.
- [8] OpenAI, *One Rational Matrix Hitting Point for Noncommutative Formulas*, manuscript dated 24 September 2026, section *A bound on the order of vanishing*, in `openai/math`. Manuscript proof; Lean source, `OAI.NCHitting.system_multiplicity`. The manuscript attributes the estimate to Moura. Sources pinned to commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, consulted 7 October 2026.
- [9] OpenAI, *Étale analytic jet lemmas*, `EtaleAnalytic.lean`, `etale_tangent_separates`, in `openai/math`, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, consulted 7 October 2026. Cited for research provenance, not as an input to the main proof.
- [10] OpenAI, *Rigidity of Smooth Billiards with a Continuous Caustic Collar*, manuscript dated 24 September 2026, section *Endpoint determination and meromorphic continuation*. Pinned section source, `openai/math`, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, consulted 7 October 2026. Cited for research provenance only; the billiard hypotheses are not asserted for the present integrals.